

University of Bahrain

College of Information Technology
Department of Computer Science

ITCS347 Design and Analysis of Algorithms

Exercises #1 – ***Writing Iterative & Recursive Algorithms***

1. Design and implement an algorithm to determine whether a given element appears anywhere before another given element in a give array $A[0..n-1]$.
 - (a) Using iterative method.
 - (b) Using recursive method.
2. Design and implement an algorithm for deleting an element e from index k in array $A[0..n-1]$.
 - (a) Design the algorithm as in-place algorithm.
 - (b) Design the algorithm as NOT in-place algorithm (i.e. use a backup array).
3. Write an algorithm that tests if an array $A[0..n-1]$ is sorted in ascending order or not.
 - (a) Write the algorithm using iterative method and find its complexity.
 - (b) Write the algorithm using recursive method and find its complexity.
4. Write an algorithm named SQUARED SUM that takes a real number x and a positive integer n as parameters. The algorithm should compute and return the sum of the following series based on the values of x and n .

$$\text{SQUARED SUM}(x, n) = \frac{x^2}{1} + \frac{x^2}{2} + \frac{x^2}{3} + \cdots + \frac{x^2}{n}$$

- (a) Write the algorithm using iterative method and find its complexity.
 - (b) Write the algorithm using recursive method and find its complexity.
5. Write a program to compute the value of $\ln(1+x)$ from the series:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \cdots$$

The algorithm should continue adding the terms to the sum with alternating signs (positive and negative), until the desired precision is achieved (e.g., the difference between successive terms is very small).

6. Write an algorithm that calculates compound interest using two different approaches: the *iterative* and *recursion* methods. The compound interest can be calculated using the following recurrence relation:

$$A_t = A_{t-1} \times \left(1 + \frac{r}{100}\right)$$

Where $A(t)$ is the amount of money accumulated after t years, including interest, r is the annual interest rate (in percentage), and t is the number of years.

7. A palindrome is a string that reads the same forward and backward, such as "radar". Write an algorithm called ISPALINDROME() that has one parameter of type string. The algorithm returns true if the argument is a palindrome and false otherwise.
 - (a) Design the algorithm using iterative method.
 - (b) Design the algorithm using recursive method.
8. Write a recursive method called DIGITMATCH() that accepts two nonnegative integers as parameters and returns the number of digits that match between them. For example, if the first number is 1072503891 and the second number is 62530841, then the number of matching digits is 4 because four of the pairs match (1-1, 8-8, 5-5, and 2-2).
 - (a) Design the algorithm using iterative method.
 - (b) Design the algorithm using recursive method.